

From black box to analytical insight: A data-driven evaluation of technological sustainability in manufacturing supply chains

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ABSTRACT

Technological infrastructures critically drive resilience and sustainability in manufacturing supply chains yet remain severely underrepresented in conventional sustainability assessment frameworks. This paper introduces the Organizational Technological Sustainability Assessment (O-TSA), an innovative data-driven model that transforms operational technology data into strategic insight. Anchored in Life Cycle Thinking, O-TSA evaluates technological sustainability through three quantifiable dimensions: Input/Output Availability, Operational Performance, and Technical Quality, each measured via weighted indicators and standardized scoring functions. The framework delivers two actionable metrics: the Technological Sustainability Index (TSI), providing a precise measurement of current technological maturity, and the Technology Improvement Index (TII), quantifying performance evolution to enable evidence-based decision-making. When applied to a ceramic tile manufacturer, the model revealed specific operational inefficiencies while documenting a significant improvement in technological sustainability over a one-year period, primarily through enhanced documentation systems and digital integration. Empirical validation confirms the model's effectiveness in converting fragmented data streams into prioritized action points. By rendering previously invisible technological dependencies explicit and measurable, the O-TSA framework enables supply chain managers to align technological investments with sustainability objectives, facilitating the development of analytically-managed, resilient industrial ecosystems in resource-intensive environments.

1. Introduction

In recent years, the notion of sustainability in manufacturing organizations has undergone a profound transformation [1]. While early interpretations were largely centered on minimizing environmental impacts and promoting economic efficiency, contemporary approaches have progressively embraced more holistic and systemic perspectives [2]. The advent of Industry 5.0 and the imminent transition to Industry 6.0 paradigms have further expanded this conceptual landscape, emphasizing human-centric innovation, resilience, and long-term value creation [3].

In this evolving context, technological systems have come to the fore as both enablers and drivers of sustainable transformation across industrial value chains [4]. The integration of advanced technologies, ranging from cyber-physical systems and Internet of Things (IoT) architectures to artificial intelligence and autonomous production, has

redefined the scope of sustainability from a merely operational concern to a strategic imperative [5]. Technologies have evolved from impartial instruments for process enhancement to dynamic agents that actively influence the sustainability trajectories of organizations.

This conceptual evolution poses a significant challenge to firms, requiring not only the deployment of innovative technologies but also the assessment of sustainability in a rigorous and structured manner [6]. In particular, the ability to monitor the technological performance of an organization over time, in relation to environmental, economic, and social goals, is regarded as a critical capability in the current era of digital-industrial convergence [7].

Despite this pivotal role, a comprehensive review of sustainability assessment literature reveals a significant analytical gap: while environmental impacts are extensively measured through standardized frameworks like Life Cycle Assessment (LCA), and economic performance through diverse financial metrics, technological sustainability

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remains largely unmeasured and unmanaged at the organizational level [8–10]. Conventional sustainability assessment models frequently prioritize material flows, energy consumption, and lifecycle environmental impacts, yet they systematically fail to capture the strategic and operational dimensions of technological resources [11].

Furthermore, when technologies are considered, they are typically framed as instruments to achieve sustainability outcomes, rather than as endogenous elements whose performance, quality, and availability warrant dedicated analysis [12]. This analytical imbalance imposes limitations on organizations' capacity to comprehensively grasp and effectively manage the intricate relationship between technology and sustainability [13].

This gap is particularly problematic for supply chain managers for three critical reasons. First, technological infrastructures represent substantial capital investments that directly impact economic sustainability [14]. Second, technological configurations determine resource efficiency, thereby influencing environmental performance [15]. Third, the resilience of supply chains, their ability to withstand disruptions, is increasingly dependent on technological adaptability and integration [16]. Without metrics to evaluate these technological dimensions, organizations lack visibility into a crucial determinant of their long-term sustainability.

In response to this gap, this article introduces and applies the Organizational Technological Sustainability Assessment (O-TSA) framework. The O-TSA model is predicated on a multidimensional vision of sustainability that explicitly incorporates the technological domain alongside environmental, economic, and social dimensions. It provides a structured methodology for evaluating technological sustainability at the organizational level.

The theoretical underpinnings of the framework have been established through prior research [17], and its operationalization entails the construction of two indices: the Technological Sustainability Index (TSI) and the Technology Improvement Index (TII). These indices facilitate analyses of technological performance that are both static and dynamic in nature. The aim of this paper is twofold. First, it seeks to advance the theoretical discourse on technological sustainability by proposing a comprehensive and integrated model for organizational-level assessment. Secondly, it exemplifies the practical applicability of the O-TSA framework through its implementation in a real-world industrial context. By undertaking this initiative, the article makes a significant contribution to the ongoing efforts to develop robust, data-driven tools that support strategic decision-making in technology-intensive manufacturing environments.

The remainder of this paper is organized as follows. Section 2 reviews the literature on sustainability assessment, decision-making frameworks, and emerging contributions on technological sustainability, highlighting the research gap that motivates the O-TSA model. Section 3 details the methodology, including research design, system mapping, indicator selection, weighting, data collection, implementation, and validation. Section 4 formalizes the O-TSA model and presents the results of its empirical application to a ceramic manufacturing case study. Section 5 discusses the theoretical and practical contributions, integration with existing sustainability frameworks, systemic implications for supply chain analytics, limitations, and future development pathways. Finally, Section 6 concludes with a summary of contributions and directions for further research.

2. Literature review

This section is structured as follows. Section 2.1 reviews the evolution of sustainability assessment in supply chains, emphasizing the environmental and economic focus of existing approaches. Section 2.2 introduces decision-making frameworks, highlighting the role of Multi-Criteria Decision-Making (MCDM) techniques. Section 2.3 identifies the limited attention given to technological factors, while Section 2.4 discusses emerging contributions on technological sustainability. Section

2.5 presents the O-TSA model as a response to these research gaps. Finally, Section 2.6 outlines the theoretical framework for the Organizational Technological Sustainability Assessment (O-TSA), which provides the conceptual basis for the methodological design presented in the next section.

2.1. Evolution of sustainability assessment in supply chains

Over the past two decades, the growing need for sustainable development has stimulated significant academic research on sustainability assessments in industrial and supply chain contexts [18]. Most contributions to this field have focused on environmental and economic aspects, paying particular attention to quantitative methodologies that facilitate structured decision-making processes and defend the competitiveness of supply chain actors [19].

Among these, Life Cycle Assessment (LCA) has emerged as a foundational tool for evaluating environmental impacts across product and process life-cycles [20]. LCA has been extensively applied to measure emissions, resource consumption, and waste generation, offering valuable insights for eco-design and sustainable manufacturing strategies [21].

Building on this foundation, Life Cycle Thinking (LCT) extends the approach to the entire value chain [22], enabling organizations to anticipate systemic consequences of decisions across stages “from cradle to grave” [23]. In this regard, LCT is well-aligned with the strategic objectives of supply chain analytics, as it facilitates a comprehensive, data-informed evaluation of performance trade-offs over time and space [16].

Although initially developed in the context of environmental analysis, LCT principles have been progressively extended to encompass economic and social dimensions, thereby providing a holistic lens for sustainability assessment in industrial networks [24].

2.2. Decision-making frameworks for sustainability

Concurrently, Multi-Criteria Decision-Making (MCDM) approaches have gained traction as flexible frameworks for integrating diverse performance indicators into coherent sustainability assessment [25]. A multitude of techniques have been utilized to address the inherent uncertainty and subjectivity of sustainability criteria.

Common techniques include the Analytic Hierarchy Process (AHP), the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [26], and fuzzy extensions of these models [27]. Recent studies have extended the application of Multi-Criteria Decision Analysis beyond operational contexts to evaluate sustainability-related dimensions such as ESG performance on social media platforms. For example, an integrated framework leveraging TOPSIS has been proposed to rank companies based on environmental, social, and governance indicators, demonstrating the versatility of MCDA in combining qualitative and quantitative criteria for sustainability assessment [28]. In the context of supply chain analytics, hybrid models that integrate fuzzy logic with MCDM have demonstrated efficacy in capturing stakeholder preferences, prioritizing sustainability factors, and structuring intricate trade-offs [29].

These approaches have enabled more nuanced evaluations of sustainability performance by accommodating the qualitative judgments of experts alongside quantitative performance metrics [30]. However, a thorough examination of these models indicates that their implementation has been predominantly focused on environmental and economic criteria, with technological dimensions typically receiving only marginal consideration [31].

2.3. The missing technological dimension in sustainability assessment

Despite the proliferation of these approaches, a critical review of the literature reveals a consistent underrepresentation of the technological

dimension within sustainability assessment models [17]. Technologies are frequently cited as enablers of more efficient and sustainable operations; however, few studies systematically evaluate the sustainability of technologies themselves, especially at the organizational level [32].

This highlights the need to treat technology not only as an enabler but as a sustainability dimension in its own right, requiring dedicated indicators and evaluative frameworks. In most cases, technological systems are regarded as a backdrop to broader evaluations of environmental performance or economic viability [33].

This framing reflects an instrumental view of technology, which fails to consider its intrinsic characteristics, such as adaptability, operational efficiency, lifecycle resilience, and technical quality [34]. These characteristics are essential to organizational sustainability over time [35].

This omission is particularly notable in prominent sustainability assessment frameworks. For instance, the Global Reporting Initiative (GRI) standards include numerous indicators for environmental and social performance but lack dedicated metrics for technological sustainability beyond basic innovation indicators [36]. In a similar vein, the United Nations Sustainable Development Goals (SDGs) acknowledge technology as a means of implementation (SDG 17) while offering minimal guidance on the evaluation of technological systems as sustainability entities [37].

This pattern is evident in industry-specific frameworks and supply chain sustainability models, where technology is instrumentalized rather than positioned as a distinct sustainability dimension.

2.4. Emerging research on technological sustainability

A paucity of studies has been dedicated to the integration of technological performance metrics into sustainability evaluation frameworks [38]. Contributions to this field have been characterized by a tendency to emphasize specific aspects, such as energy efficiency [39] or digital maturity [40], without adopting a holistic, multidimensional approach that considers the comprehensive range of technological impacts across organizational systems.

Early contributions, such as Zhang and Haapala (2012) [41] and Stock and Seliger (2016) [42], focused on equipment-level or enabling roles of technology, but stopped short of proposing comprehensive frameworks. However, their research did not extend to the development of comprehensive assessment frameworks. In their 2017 study, Badurdeen and Jawahir proposed a manufacturing process sustainability index that incorporated technological aspects, emphasizing their role as enablers of environmental performance rather than as discrete sustainability entities [43].

Additionally, there is a paucity of structured indices or models that explicitly capture the dynamics of technological sustainability, particularly with respect to monitoring improvements and identifying areas of vulnerability within a firm's technological infrastructure [44]. Existing approaches are characterized by static evaluations that fail to capture the evolutionary nature of technological systems and their changing contributions to organizational resilience over time.

2.5. The organizational technological sustainability assessment model as a response to research gaps

The Organizational Technological Sustainability Assessment (O-TSA) model aims to address this gap by offering a theoretically grounded and operationally viable framework for assessing technological sustainability within manufacturing organizations.

In contrast to conventional models that regard technology as a static enabler, O-TSA conceptualizes technology as a dynamic and assessable component of sustainability. This dynamic component is endowed with its own set of dimensions, indicators, and trajectories. Similarly to what recent studies have highlighted [45–47], showing that the synergistic combination of organizational practices is crucial to maximize production flexibility without compromising cost and quality, the O-TSA model

also emphasizes the importance of an integrated and multidimensional approach to technological sustainability, moving beyond the logic of partial evaluations. This perspective reinforces the rationale for structuring technological sustainability around interconnected dimensions, as isolated improvements rarely deliver systemic resilience or long-term value. The model builds upon a four-dimensional interpretation of sustainability (economic, environmental, social, and technological) initially proposed by Vacchi et al. [17] and advances it through the development of composite indices that quantify technological sustainability and track its evolution over time.

In doing so, the model explicitly operationalizes Life Cycle Thinking by translating the successive phases of technological systems into the three dimensions of Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ). Additionally, O-TSA draws from the systemic logic of LCT, as it is designed to capture technological impacts across different stages of organizational processes and to support longitudinal performance monitoring.

The O-TSA framework is thus positioned as a novel contribution to the field of supply chain analytics, bridging a critical gap in existing methodologies and offering a structured lens through which organizations can evaluate and improve their technological assets in line with broader sustainability goals.

The rationale for structuring technological sustainability around the three dimensions of Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ) is grounded in both systems theory and Life Cycle Thinking. IOA captures the foundational preconditions for technology deployment, including access to documentation, interfaces, and training, which correspond to the "inputs" phase of technological life-cycles.

OP reflects the efficiency and reliability of technologies in their active use phase, thereby resonating with the operational dynamics emphasized in life-cycle performance assessment. TQ, finally, addresses resilience, conformance, and adaptability, which align with the durability and end-of-life phases of technologies. This tripartite structure was selected over alternative conceptualizations (e.g., innovation maturity models or purely economic performance metrics) because it ensures a balanced coverage of resource availability, operational capability, and long-term robustness.

In this way, the O-TSA dimensions operationalize Life Cycle Thinking at the organizational level, providing a structured means of evaluating technological systems as evolving entities rather than static enablers.

The following research questions (RQs) guide this study:

- **RQ1:** *How can the technological dimension of sustainability be effectively assessed at the organizational level using a structured and multidimensional framework?*
- **RQ2:** *To what extent does the application of the O-TSA model enable manufacturing firms to identify and monitor their technological sustainability performance over time?*

2.6. Theoretical framework for the organizational technological sustainability assessment

The Organizational Technological Sustainability Assessment (O-TSA) model is predicated on an expanded conceptualization of sustainability that unambiguously incorporates technology as a fourth dimension, in addition to the well-established environmental, economic, and social dimensions. This theoretical advancement was first articulated by Vacchi et al. [17], who argued that sustainability in manufacturing systems cannot be comprehensively evaluated without a dedicated focus on the technological assets and infrastructures that underpin production processes.

In this view, technology is not merely an enabler of sustainable outcomes; rather, it is a structural component whose own sustainability must be assessed and managed over time. The O-TSA model provides a systematic framework for evaluating technological sustainability at the

organizational level, building upon this foundation.

The methodology employed to achieve this objective entails the identification and quantification of the impact of technological systems across three distinct categories. The factors under consideration are Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ). These categories collectively represent the core dimensions of technological functionality, resilience, and effectiveness within a manufacturing organization.

- **Input/Output Availability (IOA)** is a measure of the accessibility, reliability, and sufficiency of technological resources. It encompasses both material inputs, such as equipment, components, and data, and operational outputs, including production continuity and system uptime. This dimension reflects the organization's capacity to guarantee the reliable availability of its technological infrastructure, a prerequisite for sustainable operations.
- **Operational Performance (OP)** is defined as the efficiency, productivity, and responsiveness of technological systems in supporting core manufacturing processes. Indicators within this dimension are typically used to measure energy and resource efficiency, production throughput, downtime, and other metrics that reflect the operational contribution of technology to value creation.
- **The Technical Quality (TQ)** dimension is designed to evaluate the conformity, robustness, and adaptability of technological solutions over time. This dimension encompasses indicators pertaining to process precision, defect rates, ease of maintenance, and system upgradeability, thereby underscoring the significance of long-term technological reliability and innovation-readiness.

Each of these categories is operationalized through a set of quantitative indicators, selected based on three primary criteria: data availability, relevance to organizational objectives, and alignment with the principles of Life Cycle Thinking (LCT).

The incorporation of LCT into the O-TSA framework ensures that technological sustainability is evaluated with consideration for the entire life-cycle of technologies, from acquisition and implementation to obsolescence and renewal. In this regard, the model is conceptually aligned with the logic of Life Cycle Assessment (LCA) [48] and Organizational Life Cycle Assessment (O-LCA) [49], both of which emphasize systemic, longitudinal evaluation across all operational phases.

While LCA (Life Cycle Assessment) has historically concentrated on environmental impacts, and O-LCA (Organization Life Cycle Assessment) broadens the scope to encompass entire organizations, O-TSA (Organizational Technological Sustainability Assessment) is distinctive in its emphasis on technological systems as both objects and agents of sustainable transformation.

Indicators are subjected to a process of normalization and aggregation in order to construct a composite index known as the Technological Sustainability Index (TSI). The TSI functions as a synthetic measure of an organization's overall technological sustainability, enabling internal benchmarking and cross-functional comparisons.

The model's incorporation of a Technology Improvement Index (TII) serves to complement the static perspective offered by the TSI. The TII has been designed to track the evolution of technological sustainability over time. The TII is an indicator that quantifies relative improvements (or regressions) in the TSI by comparing values across successive assessment periods. This dynamic component enhances the model's utility for strategic planning and continuous improvement, enabling decision-makers to monitor technological trajectories and identify early signals of decline or advancement.

Therefore, the O-TSA framework is a structured approach to assess the sustainability of technological systems at the organizational level. By aligning technological assessment with established sustainability paradigms, O-TSA highlights potential synergies and trade-offs with environmental and social goals, ensuring that technological infrastructures are evaluated not in isolation but as integral components of systemic

sustainability.

The model is founded on the principles of Life Cycle Thinking and is built upon a multidimensional understanding of sustainability. It defines clear categories and indicators that reflect the operational, systemic, and quality-related impacts of technology. This conceptual framework provides the foundation for its empirical implementation and validation within industrial contexts.

3. Methods

This section is structured as follows. Section 3.1 outlines the research design and epistemological stance. Section 3.2 describes the mapping of technological systems and scope definition, followed by the indicator selection and validation process in Section 3.3. Section 3.4 details the weight assignment methodology, and Section 3.5 explains data collection and triangulation. Section 3.6 illustrates the implementation of the O-TSA model, while Section 3.7 concludes with the validation approach.

3.1. Research design and epistemological approach

The development and implementation of the Organizational Technological Sustainability Assessment (O-TSA) framework are rooted in a constructivist epistemological approach, wherein the model is not merely tested but also iteratively refined through interaction with real-world organizational contexts [50]. In contradistinction to designs that are purely positivist in nature, which assume a clear separation between subject and object, the constructive perspective acknowledges the active role of the researcher in shaping the analytical model in response to the complexity of industrial systems. This orientation is particularly well-suited to emerging phenomena such as technological sustainability, where standardized metrics and evaluation tools are still in a state of development.

From a methodological perspective, the research design employs design science logic with the objective of constructing and validating a structured framework for the evaluation of technological sustainability within manufacturing organizations. The O-TSA model is applied in a single-case empirical study [51] within a representative industrial firm in the ceramic sector, a domain characterized by high resource and energy intensity, significant technological dependency, process automation, and complexity in the supply chain [32]. The selected case offers a valuable opportunity to explore the multidimensional nature of technological sustainability and to assess the operational viability of the proposed indicators and indices.

3.2. Technological system mapping and scope definition

The assessment began with a detailed mapping of the technological systems present within the various areas of the organization. In this preliminary phase, the main technological subsystems along the entire value chain, from raw material processing to finished product logistics, were precisely identified. Each technology was classified according to its function (transformation, control, integration, or support) and all interfaces and dependencies among subsystems were carefully documented. At the same time, the availability of data necessary for analysis was verified for each technological component.

This systematic approach made it possible to clearly define the scope of the assessment and to identify the key technological assets on which to focus further analysis. Deliberate attention was given to production technologies with the greatest impact in terms of resources and strategic importance, while peripheral systems with limited influence on sustainability were excluded from the scope of analysis.

At the same time, the mapping process underscored the importance of aligning the O-TSA with broader digital transformation initiatives. The framework is inherently compatible with data streams generated by IoT-enabled monitoring devices, digital twin simulations, and AI-driven analytics platforms. This compatibility ensures that, as organizations

progress in their digitalization journey, the O-TSA can leverage increasingly granular and real-time data, thereby enhancing the timeliness and accuracy of sustainability insights.

3.3. Indicator selection and validation process

Following the mapping of technological systems, a structured process was undertaken to select appropriate indicators for each dimension of technological sustainability. Initially, a comprehensive pool of thirty-five potential indicators was developed, drawing on an extensive literature review, industry best practices, and consultation with technical experts. This initial set encompassed a broad range of sources, including manufacturing performance literature, equipment effectiveness metrics, and established technology assessment frameworks.

The pool was then rigorously filtered using three primary criteria: the relevance of each indicator to technological sustainability in the organizational context, the feasibility of consistently measuring each indicator with existing data systems, and the indicator's analytical value with respect to supporting strategic decision-making and improvement planning.

In this stage, data availability played a decisive role. Several theoretically relevant indicators could not be included because they were not supported by reliable or consistent measurements in the company's enterprise resource planning (ERP) and monitoring systems. In such cases, the indicator set was pragmatically adjusted, and thresholds were calibrated on external benchmarks whenever internal historical data were insufficient. This ensured operational feasibility but also introduced some dependency on sectoral reference values.

Expert evaluation and consensus-based discussion further refined the selection process. Consensus was built through structured workshops involving managers from production, maintenance, and quality departments. Divergent opinions on indicator relevance or threshold calibration were reconciled using facilitated discussions, where preliminary results were compared against both organizational priorities and external benchmarks. This iterative dialogue allowed participants to converge on shared definitions and weightings, strengthening the reliability of the final indicator set.

Ultimately, a final set of indicators was identified for each of the model's three key dimensions: Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ). This methodical approach ensured that the resulting indicator set was both theoretically robust and practically applicable within the organization, striking a balance between conceptual soundness and operational feasibility.

3.4. Weight assignment methodology

The weighting system for the O-TSA model was developed through a combination of analytical and participatory approaches designed to ensure both methodological rigor and organizational validity.

- i. **Initial weights through pairwise comparison:** An adapted Analytic Hierarchy Process (AHP) was employed to establish baseline weights for each indicator within its respective dimension. Technical managers from production, maintenance, and quality departments performed pairwise comparisons, rating the relative importance of indicators on a 1–9 scale.
- ii. **Consistency analysis:** For each comparison matrix, the consistency ratio (CR) was calculated to verify logical coherence. Whenever the CR exceeded the recommended threshold of 0.10, the judgments were reviewed in a dedicated round of discussion and the comparisons repeated until consistency was achieved.
- iii. **Sensitivity testing:** Multiple weight configurations were tested to assess the robustness of the model to changes in weight distribution. Thresholds were identified beyond which modifications would materially affect the aggregated assessment, thereby

ensuring that the results were not overly sensitive to small variations in weights.

- iv. **Consensus refinement:** A facilitated workshop was organized to consolidate the individual judgments. During this session, preliminary weight sets were presented, and participants engaged in structured discussion to reconcile divergent opinions. When substantial differences persisted, participants were invited to justify their judgments, after which a shared evaluation was negotiated. The CR was recalculated following each adjustment to confirm logical consistency.

This multi-step procedure combined the transparency of expert participation with the robustness of mathematical checks, ensuring that final weights reflected both technical importance and stakeholder priorities while remaining consistent and reproducible. Recent advances in hybrid Multi-Criteria Decision Analysis approaches have demonstrated the potential to enhance consistency and reliability in ranking and weighting processes. For instance, [52] introduced a Best-Worst Method combined with Data Envelopment Analysis to achieve full ranking of decision-making units, reducing subjectivity and improving robustness compared to traditional techniques. This perspective reinforces the relevance of structured weighting systems within the O-TSA framework.

To strengthen robustness, additional sensitivity analyses were performed by varying indicator weights within plausible ranges (± 10 – 20 %) and observing the resulting variation in the Technological Sustainability Index (TSI). These tests showed that while minor adjustments had limited influence on the final outcome, more extreme re-distributions shifted dimension-level balances without altering the overall ranking of technological strengths and weaknesses. This analysis confirms that the framework's diagnostic value remains stable across reasonable variations in weight assignment.

Looking ahead, greater objectivity in weight assignment could be achieved through automated procedures such as entropy weighting, machine learning algorithms for feature importance, or hybrid approaches that integrate expert judgment with data-driven calibration. These methods would reduce reliance on subjective assessments while preserving contextual validity, thereby enhancing replicability across organizational settings.

3.5. Data collection and triangulation

The data collection process entailed a multifaceted approach, encompassing both primary and secondary sources. Structured interviews and internal workshops were carried out with managers and technical personnel from production, quality, and maintenance departments. Concurrently, extensive documentation was retrieved from ERP systems, sustainability reports, and key performance indicator (KPI) dashboards. Specific data sources included:

- Production monitoring systems and Supervisory Control and Data Acquisition (SCADA) platforms (for real-time operational metrics),
- Maintenance management systems (for reliability and availability data),
- Quality control databases (for conformance and defect information),
- Energy management systems (for consumption patterns and efficiency metrics),
- Technical documentation repositories (for system specifications and update histories).

Data triangulation was achieved by cross-validating information across multiple sources and through iterative verification with technical staff. This approach ensured data integrity while mitigating potential biases from any single information source.

More broadly, implementing the O-TSA requires careful attention to data governance and system integration. Successful application depends not only on extracting data from Enterprise Resource Planning (ERP),

Manufacturing Execution System (MES), and Supervisory Control and Data Acquisition (SCADA) systems but also on ensuring consistency across these platforms through standardized interfaces and metadata structures. Organizations with fragmented digital architectures may need to invest in middleware solutions or data lakes to harmonize inputs. Equally important are data quality assurance mechanisms, including automated validation routines and version control, which safeguard the reliability of longitudinal assessments and enable interoperability with advanced analytics tools. By clarifying these requirements, the O-TSA framework positions itself as a bridge between sustainability assessment and the ongoing digitalization of supply chain operations.

3.6. Model implementation process

The implementation of the O-TSA framework proceeded through three primary phases:

- i. **Indicator measurement and scoring:** Raw indicator values were collected and converted to standardized scores using predefined threshold functions. These functions mapped performance levels to a 1–5 scale based on industry benchmarks, historical performance, and technical specifications.
- ii. **Dimensional aggregation:** Indicator scores were combined using the weighted aggregation formulas described in Section 5 to produce dimensional indices for Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ).
- iii. **Index calculation and temporal comparison:** The Technological Sustainability Index (TSI) was computed as a composite measure of overall technological sustainability, while the Technology Improvement Index (TII) was calculated to evaluate performance evolution between assessment periods.

The analytical process was conducted using a combination of spreadsheet-based modeling and statistical analysis tools. While fuzzy logic or other advanced MCDM techniques were not employed in this initial application, the structure of the model remains compatible with such methods for future developments. The current implementation prioritizes clarity, replicability, and ease of use in operational settings, in line with the framework's intended role as a decision-support tool within manufacturing organizations.

3.7. Validation approach

The validity of the O-TSA model was established through multiple mechanisms:

1. **Content validity:** The comprehensiveness and relevance of the model dimensions and indicators were validated through expert review and alignment with established technological management principles.
2. **Construct validity:** The internal consistency of the model was assessed through correlation analysis between indicators and dimensions to verify logical relationships.
3. **Predictive validity:** The model's results were compared with independent technological performance metrics to assess alignment with observed operational outcomes.
4. **Practical validity:** The utility and actionability of the model outputs were evaluated through structured feedback from organizational decision-makers following implementation.

This validation approach ensured that the O-TSA model not only possessed theoretical soundness but also demonstrated practical value in the organizational context, fulfilling its dual purpose as an analytical framework and a decision-support tool.

4. Results

This section is structured as follows. Section 4.1 presents the formalization of the O-TSA model, detailing its analytical structure and computational logic. Section 4.2 reports the results of the empirical case study, including the assessment of the three technological sustainability dimensions, the computation of the Technological Sustainability Index (TSI) and the Technology Improvement Index (TII), and their strategic interpretation within the supply chain context.

4.1. Model formalization and analytics structure

The Organizational Technological Sustainability Assessment (O-TSA) model is a hierarchical, multi-criteria framework designed to evaluate the technological sustainability of an organization based on a structured set of indicators grouped into three key dimensions. The model employs a semi-quantitative approach, using discrete scoring, predefined performance thresholds, and weighted aggregation to compute a synthetic index of technological sustainability, the TSI, and its temporal evolution, the TII.

4.1.1. Structural definition

To formalize the Organizational Technological Sustainability Assessment (O-TSA) model, the following mathematical notation is introduced. The formulation defines the key dimensions, indicators, and parameters used to quantify technological sustainability and to construct the Technological Sustainability Index (TSI) and the Technology Improvement Index (TII).

Let:

$D = \{1, 2, 3\}$ be the set of technological sustainability dimensions, corresponding to:

- $d = 1$: Input/Output Availability (IOA)
- $d = 2$: Operational Performance (OP)
- $d = 3$: Technical Quality (TQ)

For each dimension $d \in D$, define:

- $I_d = \{1, 2, \dots, n_d\}$ the set of indicators associated with dimension d
- $x_{id} \in \mathbb{R}$ the measured value of indicator $i \in I_d$
- $s_{id} \in \{1, 2, 3, 4, 5\}$ the assigned score based on the value of x_{id} , using threshold-based classes
- $w_{id} \in [0, 1]$ the weight of indicator i in dimension d , such that $\sum_{i \in I_d} w_{id} = 1$

Each indicator is evaluated through a performance function $f_{id}(x_{id})$, which maps the raw measurement to a discrete score according to predefined thresholds:

$$s_{id} = f_{id}(x_{id}) = \begin{cases} 1 & \text{if } x_{id} < \tau_1^{(id)} \\ 2 & \text{if } \tau_1^{(id)} \leq x_{id} < \tau_2^{(id)} \\ 3 & \text{if } \tau_2^{(id)} \leq x_{id} < \tau_3^{(id)} \\ 4 & \text{if } \tau_3^{(id)} \leq x_{id} < \tau_4^{(id)} \\ 5 & \text{if } x_{id} \geq \tau_4^{(id)} \end{cases} \quad (1)$$

where $\tau_j^{(id)}$ are the thresholds (class boundaries) for the indicator i in dimension d , defined a priori based on technological standards, best practices, or expert judgment.

Thresholds were calibrated by triangulating three sources: international standards and industry benchmarks (e.g., ISO 14040/44 for life-cycle performance, ISO 55000 for asset management), historical operational data from the company, and expert judgment obtained during workshops. When discrepancies arose across sources, priority was given

to values that were both measurable in practice and recognized as realistic by internal experts, ensuring contextual validity of the scoring system.

To facilitate replication, the calibration process followed a three-step protocol: (i) external benchmarks and international standards were collected and normalized to a 1–5 scoring scale; (ii) internal performance data were analyzed to identify historical distributions and feasible thresholds; (iii) expert panels validated the proposed thresholds against operational relevance. This structured approach enables adaptation across sectors while ensuring consistency and transparency in threshold setting.

In future applications, threshold calibration could be increasingly automated. Data-driven approaches such as statistical distribution analysis, clustering algorithms, or supervised learning models have the potential to identify class boundaries directly from large datasets. Incorporating such methods would reduce subjective influence in threshold definition and align the scoring functions more closely with empirical performance patterns observed across industries.

4.1.2. Dimensional aggregation

Each dimension-level score S_d is computed as the weighted average of the scores of its constituent indicators:

$$S_d = \sum_{i \in I_d} w_{id} \bullet s_{id} \quad (2)$$

This operation yields a normalized dimensional score in the range [1, 5], representing the overall technological sustainability of the organization in that particular dimension.

Let:

$$S = (S_1, S_2, S_3) \in \mathbb{R}^3 \quad (3)$$

be the dimensional sustainability vector.

4.1.3. Computation of the technological sustainability index

To obtain a synthetic measure of the organization's overall technological sustainability, the dimensional scores are aggregated using another set of weights $\alpha = (\alpha_1, \alpha_2, \alpha_3) \in [0, 1]^3$, such that $\sum_{d=1}^3 \alpha_d = 1$

The **Technological Sustainability Index** (TSI) is given by:

$$TSI = \sum_{d=1}^3 \alpha_d \bullet S_d \quad (4)$$

This index assumes values in the range [1, 5] and captures the technological sustainability level of the organization at a given point in time.

Special case – Equal weighting

If no preferential weights are defined, we assume

$$\alpha_d = \frac{1}{3} \text{ for all } d \in D \quad (5)$$

yielding a simple arithmetic mean of the three-dimension scores.

4.1.4. Temporal assessment: technology improvement index

To evaluate the evolution of technological sustainability over time, the Technology Improvement Index (TII) measures the relative variation of TSI between two periods t and $t + 1$:

$$TII = \frac{TSI^{(t+1)} - TSI^{(t)}}{TSI^{(t)}} \bullet 100 \quad (6)$$

where:

- $TSI^{(t)}$: technological sustainability index at time t
- $TSI^{(t+1)}$: index at a later period

Interpretation:

- $TII > 0$: technological improvement
- $TII = 0$: stability
- $TII < 0$: technological decline

4.1.5. Matrix Formulation

Define:

- $W_d = [w_{1d}, w_{2d}, \dots, w_{n_d d}]$ - vector of weights for dimension d
- $S_d = [s_{1d}, s_{2d}, \dots, s_{n_d d}]$ - vector of scores

Then:

$$S_d = W_d \bullet S_d^T \quad (7)$$

Let:

- $\alpha = [\alpha_1, \alpha_2, \alpha_3]$
- $S = [S_1, S_2, S_3]$

Then:

$$TSI = \alpha \bullet S^T \quad (8)$$

This matrix formulation facilitates the use of vector-based computation in spreadsheet or scripting environments (e.g., Python with NumPy, R, MATLAB).

4.1.6. Visual representation of the analytical model

In order to provide a more comprehensive understanding of the OTSA framework, this subsection introduces two visual representations that elucidate the internal logic and computational structure of the model. These visual elements serve to augment the model's transparency and operational usability, a benefit that is particularly pronounced for practitioners and analysts who are implementing the framework within organizational contexts.

Table 1 presents the detailed matrix used for the computation of the Technological Sustainability Index (TSI). It includes the three technological sustainability dimensions—Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ)—along with their respective indicators, assigned weights, and performance scores. The weighted contribution of each indicator is reported in the last

Table 1

Organizational Technological Sustainability Assessment weighted score matrix and aggregation.

Dimension	Indicator	Weight (w_{ij})	Score (s_{ij})*	$w_{ij} \bullet s_{ij}$
Input/Output Availability	x_{11}	0.3	4	1.20
	x_{12}	0.4	3	1.20
	x_{13}	0.3	5	1.50
Operational Performance	x_{21}	0.3	3	0.90
	x_{22}	0.3	4	1.20
	x_{23}	0.4	2	0.80
Technical Quality	x_{31}	0.5	5	2.50
	x_{32}	0.5	4	2.00
			S_d	
Input/Output Availability				3.90
Operational Performance				2.90
Technical Quality				4.50
			Technological Sustainability Index	3.77

* Score: 1 = lowest; 5 = highest.

column, while dimension-level aggregates are obtained as the weighted sum of individual scores. The overall Technological Sustainability Index is then calculated as the average of the three dimensions.

The data reported in Table 1 derives from a hypothetical application of the O-TSA model using synthetic values constructed for illustrative purposes. The performance scores s_{ij} are assigned based on predefined threshold classes (see Section 4.4.1), which map raw indicator values x_{ij} into discrete scores ranging from 1 (lowest performance) to 5 (highest performance). While the raw values are not shown here, each score corresponds to a specific class according to a standardized scoring function $f_{ij}(x_{ij})$ calibrated on technological benchmarks or internal operational targets.

The weights w_{ij} represent the relative importance of each indicator within its corresponding dimension. In this illustrative case, weights are assigned arbitrarily while respecting the normalization condition $\sum_i w_{ij} = 1$ within each dimension. In operational settings, these weights can be defined through expert judgment, stakeholder consultation, or data-driven techniques such as the Analytic Hierarchy Process (AHP) or entropy weighting.

In this study, weights were established through an AHP-based procedure complemented by a structured consensus workshop. Technical managers from production, maintenance, and quality first provided independent pairwise comparisons, which were consolidated and discussed collectively. Divergent judgments were reconciled through facilitated dialogue until convergence was reached, and the consistency ratio (CR) was recalculated after each adjustment to confirm logical coherence. This participatory process ensured that the final weighting system reflected both analytical rigor and stakeholder consensus.

The dimensional scores S_d are computed as the weighted average of the corresponding s_{ij} values, and the Technological Sustainability Index (TSI) is obtained by aggregating the S_d values using equal weights across the three dimensions. This structured process ensures consistency, comparability, and flexibility in adapting the model to different data contexts.

Each row corresponds to a specific indicator x_{ij} , grouped under one of the three dimensions: Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ). For each indicator, the assigned weight w_{ij} , the corresponding performance score s_{ij} , and the weighted contribution $w_{ij} \cdot s_{ij}$ are reported. Dimensional aggregates S_d are then calculated as the weighted average of the scores in each group. The final row of the table displays the TSI as the weighted average of the dimensional scores. This matrix structure provides a clear and modular overview of how each indicator influences the overall assessment and allows for easy adaptation when applying the model to different organizational settings.

Fig. 1 provides a schematic overview of the logical and computational flow underlying the O-TSA framework. The diagram is structured in sequential stages, each corresponding to a distinct step in the assessment process. This visual representation clarifies the modularity and hierarchical structure of the model, making it easier to understand

and implement.

The process begins with the input of raw data x_{ij} , collected from technological systems, ERP databases, technical documentation, or process monitoring platforms. These values are not directly aggregated but first undergo a scoring transformation via a performance function $f_{ij}(x_{ij})$, which assigns a discrete score $s_{ij} \in \{1, 2, 3, 4, 5\}$ based on predefined threshold classes, as defined in Section 4.1.1. Concurrently, each indicator is assigned a weight w_{ij} reflecting its relative importance within the corresponding dimension.

These weights are then used to calculate the weighted scores $w_{ij} \cdot s_{ij}$, which represent the elementary contributions of each indicator to the assessment. In the next stage, the weighted scores are aggregated by dimension (IOA, OP, TQ) to obtain the dimensional scores S_d , which summarize the organization's performance in each area of technological sustainability. Finally, the dimensional scores are aggregated globally, using a predefined set of weights S_d , to compute the Technological Sustainability Index (TSI). This index provides a synthetic view of the organization's technological sustainability at a given time. If a temporal comparison is conducted, the model also calculates the Technology Improvement Index (TII) as a percentage variation in the TSI between two periods. This modular flow ensures that the O-TSA model remains flexible, transparent, and applicable across different data environments and organizational contexts.

The analytical structure outlined above provides the methodological foundation for the empirical application of the O-TSA framework, which is presented in detail in the following section through a case study from the ceramic manufacturing industry.

4.2. Case study results

The case study serves to validate the applicability of the O-TSA model in a real-world industrial context, demonstrating how the framework can be used to quantify, interpret, and improve technological sustainability at the organizational level. This section presents the comprehensive application of the Organizational Technological Sustainability Assessment (O-TSA) model to a manufacturing company in the ceramic industry. Each phase of the assessment is described in detail, including system mapping, indicator scoring, weighted aggregation, and temporal comparison.

4.2.1. Organizational and technological context

The case study was conducted in a ceramic tile manufacturing facility that operates within a value chain that is characterized by significant resource consumption and energy intensity. The company's organizational structure is characterized by a vertically integrated production process, which encompasses the phases delineated in Fig. 2.

Each of these phases is supported by specific technological systems (mechanical, thermal, electronic, and digital) which collectively form the technological infrastructure of the plant. For the purpose of the O-TSA, the assessment focused on a subset of high-impact technological

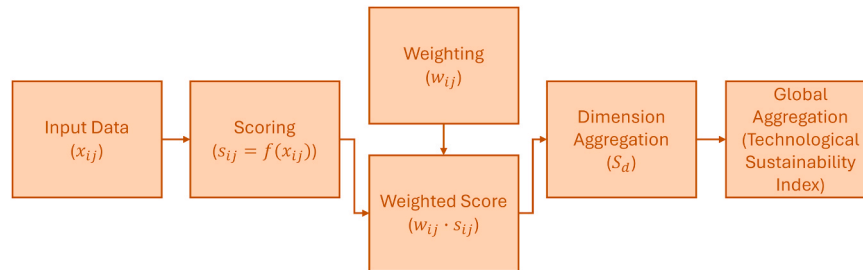


Fig. 1. Organizational Technological Sustainability Assessment model: analytical flow structure. The diagram summarizes the logical and computational process of the Organizational Technological Sustainability Assessment. It illustrates how raw operational data are transformed into standardized scores, aggregated by indicator and dimension, and then synthesized into the Technological Sustainability Index. This step-by-step visualization clarifies the modular design of the framework and its applicability to different organizational contexts.



Fig. 2. Ceramic tile production process (Elaborated from [53]). This schematic representation outlines the main phases of the ceramic manufacturing process, from raw material preparation to the final stages of tile finishing.

subsystems in the upstream and midstream stages: feeding and dosing systems, milling and mixing systems, atomization units, pressing and drying processes, and digital control systems.

4.2.2. Methodological approach and data sources

The implementation of the O-TSA model followed a structured and iterative methodological process. The model was adapted to the specific organizational context through a combination of quantitative analysis, field engagement, and expert validation. The process began with thorough mapping of the technological infrastructure, with emphasis on core subsystems critical for production. A set of performance indicators was identified for each of the three dimensions of technological sustainability: Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ).

Each indicator was x_{ij} associated with a scoring function $f_{ij}(x_{ij})$, which mapped raw measurements into discrete scores $s_{ij} \in \{1, 2, 3, 4, 5\}$. The scoring logic utilized threshold-based classes calibrated on best practices, industry benchmarks, and internal performance targets. Each indicator was assigned a weight $w_{ij} \in [0, 1]$, representing its importance within the respective dimension. These weights were determined through structured stakeholder workshops and ensured the normalization condition $\sum_i w_{ij} = 1$ was satisfied within each dimension. Data collection employed triangulation across multiple sources, including internal technical documentation, maintenance and quality records, ERP exports, and structured interviews with technical staff, reinforcing the validity of the assessment results.

4.2.3. Dimensional analysis

The implementation of the O-TSA model is articulated across three core dimensions of technological sustainability. Each dimension was evaluated through specific indicators, measured and scored according to the methodology described. For the Input/Output Availability (IOA) dimension, Table 2 reports the unit of measurement, assigned weight, scoring thresholds, observed value, and corresponding discrete score for

each indicator. This dimension reflects the organization's capacity to ensure adequate documentation, preventive maintenance, system integration, and training resources, key enablers of technological readiness and long-term reliability.

The organization demonstrated an advanced level of technological resource availability with a dimension score calculated as:

$$\begin{aligned} S_{IOA} &= (0.30 \cdot 5) + (0.30 \cdot 5) + (0.25 \cdot 4) + (0.15 \cdot 4) \\ &= 4.65 \end{aligned} \quad (9)$$

For the Operational Performance (OP) dimension, Table 3 presents the indicators evaluating efficiency and reliability of technological systems. Each indicator is defined in terms of measurement unit, assigned weight, threshold classes, observed value, and corresponding score. Collectively, these indicators assess key aspects such as energy consumption, mean time between failures, downtime, and process variability, providing a quantitative view of the operational robustness and efficiency of the technological infrastructure.

The dimension score was calculated as:

$$\begin{aligned} S_{OP} &= (0.30 \cdot 4) + (0.30 \cdot 3) + (0.25 \cdot 3) + (0.15 \cdot 3) \\ &= 3.30 \end{aligned} \quad (10)$$

For the Technical Quality (TQ) dimension, Table 4 presents the indicators concerning resilience, compliance, and flexibility of technologies. Each indicator includes its measurement unit, assigned weight, threshold classes, observed value, and corresponding score. This dimension reflects the robustness and flexibility of the technological infrastructure, capturing the organization's capacity to maintain conformance, ensure system reliability, and adapt to evolving operational requirements.

The dimension score was calculated as:

$$S_{TQ} = (0.35 \cdot 4) + (0.40 \cdot 4) + (0.25 \cdot 3) = 3.85 \quad (11)$$

Table 2

Structure of the indicators used to assess the Input/Output Availability dimension.

Indicator	Unit	w_{ij}	Threshold classes (Score 1–5)*	Observed value	Assigned score
Availability of technical documentation	% complete	0.30	$x \leq 70 \rightarrow 1$ $70 < x \leq 80 \rightarrow 2$ $80 < x \leq 90 \rightarrow 3$ $90 < x \leq 95 \rightarrow 4$ $x > 95 \rightarrow 5$	98	5
Preventive maintenance coverage	% assets covered	0.30	$x \leq 50 \rightarrow 1$ $50 < x \leq 60 \rightarrow 2$ $60 < x \leq 75 \rightarrow 3$ $75 < x \leq 90 \rightarrow 4$ $x > 90 \rightarrow 5$	91	5
Availability of system interfaces	% connected systems	0.25	$x \leq 40 \rightarrow 1$ $40 < x \leq 60 \rightarrow 2$ $60 < x \leq 75 \rightarrow 3$ $75 < x \leq 90 \rightarrow 4$ $x > 90 \rightarrow 5$	85	4
Access to training resources	days/year	0.15	$x \leq 1 \rightarrow 1$ $x = 2 \rightarrow 2$ $x = 3 \rightarrow 3$ $x = 4 \rightarrow 4$ $x > 4 \rightarrow 5$	4	4

* Score: 1 = lowest; 5 = highest.

Table 3

Structure of the indicators used to assess the Operational Performance (OP) dimension.

Indicator	Unit	w_{ij}	Threshold classes (Score 1–5)*	Observed value	Assigned score
Energy consumption	kWh/kg	0.30	$x \geq 0.35 \rightarrow 1$ $0.30 \leq x < 0.35 \rightarrow 2$ $0.27 \leq x < 0.30 \rightarrow 3$ $0.24 \leq x < 0.27 \rightarrow 4$ $x < 0.24 \rightarrow 5$	0.26	4
Mean Time Between Failures	hours	0.30	$x \leq 10 \rightarrow 1$ $10 < x \leq 20 \rightarrow 2$ $20 < x \leq 35 \rightarrow 3$ $35 < x \leq 50 \rightarrow 4$ $x > 50 \rightarrow 5$	25	3
Unplanned downtime	hours/month	0.25	$x \geq 10 \rightarrow 1$ $8 \leq x < 10 \rightarrow 2$ $5 \leq x < 8 \rightarrow 3$ $3 \leq x < 5 \rightarrow 4$ $x < 3 \rightarrow 5$	6	3
Process cycle time variation	%	0.15	$x \geq 10 \rightarrow 1$ $8 \leq x < 10 \rightarrow 2$ $5 \leq x < 8 \rightarrow 3$ $2 \leq x < 5 \rightarrow 4$ $x < 2 \rightarrow 5$	5	3

* Score: 1 = lowest; 5 = highest.

Table 4

Structure of the indicators used to assess the Technical Quality (TQ) dimension.

Indicator	Unit	w_{ij}	Threshold classes (Score 1–5)*	Observed value	Assigned score
Remaining Useful Life	years	0.35	$x \leq 3 \rightarrow 1$ $3 < x \leq 6 \rightarrow 2$ $6 < x \leq 8 \rightarrow 3$ $8 < x \leq 10 \rightarrow 4$ $x > 10 \rightarrow 5$	9	4
Conformance rate	% within spec	0.40	$x \leq 85 \rightarrow 1$ $85 < x \leq 90 \rightarrow 2$ $90 < x \leq 95 \rightarrow 3$ $95 < x \leq 99 \rightarrow 4$ $x > 99 \rightarrow 5$	97 %	4
Changeover adaptability	min/format	0.25	$x \geq 45 \rightarrow 1$ $35 \leq x < 45 \rightarrow 2$ $25 \leq x < 35 \rightarrow 3$ $15 \leq x < 25 \rightarrow 4$ $x < 15 \rightarrow 5$	25	3

* Score: 1 = lowest; 5 = highest.

The dimensional analysis revealed a balanced technological profile, with Input/Output Availability (IOA) emerging as the strongest dimension (score 4.65), followed by Technical Quality (TQ) at 3.85, and Operational Performance (OP) at 3.30.

4.2.4. Aggregated assessment and strategic interpretation

The culminating phase entailed the consolidation of the three-dimensional scores into the unified Technological Sustainability Index (TSI). With equal weights assigned to the three dimensions, the global TSI was computed as:

$$TSI = \frac{1}{3}(S_{IOA} + S_{OP} + S_{TQ}) = \frac{1}{3}(4.65 + 3.30 + 3.85) = 3.93 \quad (12)$$

This value suggests a medium-high level of technological sustainability. The score indicates robust capacity in the management of technological resources and documentation (IOA), substantial technical quality infrastructure (TQ), and moderate operational efficiency (OP), which is identified as the primary domain for enhancement.

The relatively lower score in Operational Performance (3.30) highlights a critical area where inefficiencies directly translate into higher energy consumption, increased maintenance costs, and reduced throughput. For management, these weaknesses represent both risks and opportunities: improving OP could lower energy expenditures, enhance production stability, and strengthen the firm's competitive positioning in an industry where cost efficiency and reliability are decisive factors. In this sense, the O-TSA not only identifies performance gaps but also signals their strategic implications, providing a clear roadmap for prioritizing investments in process optimization, energy-efficient technologies, and advanced maintenance practices.

To enrich the analysis, the TSI value was compared to the result of a previous assessment conducted twelve months earlier, which had yielded a TSI of 3.48. The longitudinal variation was captured using the Technology Improvement Index (TII):

$$TII = \frac{3.93 - 3.48}{3.48} \cdot 100 = 12.93\% \quad (13)$$

This 12.93 % increase in TSI over one year reflects not just incremental adjustments but structured interventions implemented at the organizational level, including enhanced documentation integration, redesigned maintenance scheduling, improved digital interfaces, and targeted training programs.

More specifically, the integration of technical documentation was supported by the creation of a centralized digital repository accessible across departments, which reduced information bottlenecks and improved process traceability. The redesign of maintenance scheduling introduced predictive maintenance practices based on condition-monitoring data, reducing unplanned downtime and extending equipment availability. Improved digital interfaces facilitated interoperability between legacy systems and newly adopted platforms, enabling more reliable data flows and decision-making support. Targeted training programs were delivered not only to technical staff but also to line operators, fostering greater user acceptance of new digital tools and ensuring effective skill transfer. These interventions collectively underpin the observed performance improvements and illustrate the types of organizational change initiatives that can translate analytical insights into tangible sustainability gains.

A workshop with operations, maintenance, and quality assurance teams translated these results into actionable insights, establishing a correlation between observed weaknesses and strategic priorities. The organization subsequently implemented retrofitting of milling line motors to enhance energy efficiency, standardized interfaces across

equipment types, and formally adopted the TSI as a key performance indicator in the company's digital transition roadmap. Fig. 3 visually synthesizes the dimensional scores, highlighting the relative strengths and improvement opportunities across the organization's technological systems.

To enhance interpretability, visual representations of temporal trends were developed. In particular, a line chart depicting the year-to-year evolution of the Technological Sustainability Index (TSI) and its dimensional components (IOA, OP, TQ) was employed to illustrate how specific improvements translated into the overall increase captured by the TII. This visualization provided a clear and intuitive representation of performance dynamics over time, making the framework's diagnostic and monitoring capacity more accessible to practitioners.

Beyond static representations, it is recommended to adopt interactive dashboards or dynamic visualizations (e.g., animated radar charts or drill-down interfaces), as these can significantly enhance practitioner engagement by enabling real-time exploration of dimensional improvements and their contribution to overall sustainability trajectories.

4.2.5. Embedding technological sustainability in the supply chain context

The assessment of technological sustainability is inextricably linked to the broader supply chain configuration. As illustrated in Fig. 5, the ceramic supply chain comprises structured production and logistics phases, each corresponding to specific technological requirements.

Within this supply chain architecture, the O-TSA model functions as a meso-level analytical framework, connecting the operational depth of technology with the strategic breadth of supply chain interdependencies. Unlike product-level or process-level evaluations that emphasize isolated units of analysis, the O-TSA model encompasses the collective technological capabilities of the entire organization. The O-TSA model employs a systemic perspective to examine how an organization as a whole manages the complexity of its technological systems to deliver performance, adaptability, and resilience over time. This characteristic renders it especially well-suited to strategic decision-making processes, particularly within industries where technological inertia and innovation gaps can compromise long-term competitiveness.

5. Discussion

This section is structured as follows. Section 5.1 highlights the theoretical and practical contributions of the O-TSA model. Section 5.2 discusses its integration with existing sustainability frameworks, while Section 5.3 elaborates the implications for supply chain analytics and digital transformation. Section 5.4 links the O-TSA framework to prescriptive analytics and systemic resilience, showing how organizational

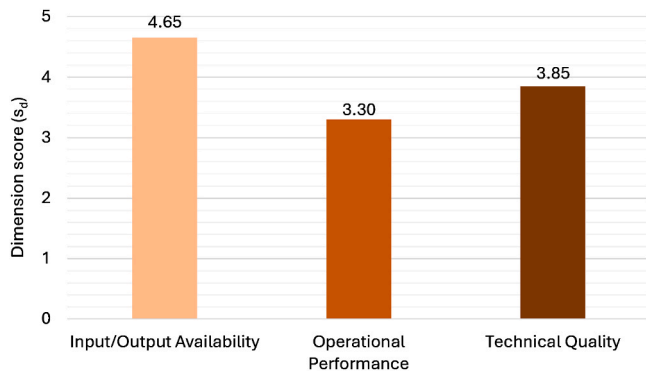


Fig. 3. Dimensional Scores of Technological Sustainability. This figure compares the performance of the three sustainability dimensions (Input/Output Availability, Operational Performance, and Technical Quality) based on the weighted scores calculated in the case study. The visualization highlights Input/Output Availability as the strongest dimension, Technical Quality as moderately strong, and Operational Performance as the main area for improvement.

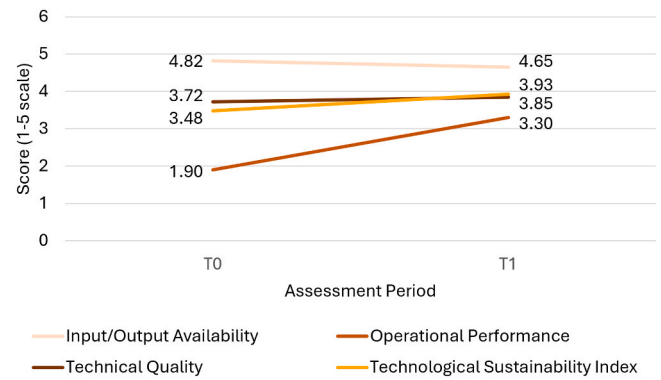


Fig. 4. Temporal evolution of technological sustainability performance. The chart illustrates how improvements in specific dimensions (notably operational performance) contributed to the observed 12.93 % increase in the Technological Sustainability Index over a one-year period. This dynamic perspective complements the radar chart by highlighting performance trajectories rather than static scores.

technology metrics can inform optimization rules, policy design, and cross-firm governance mechanisms. Section 5.5 critically reflects on limitations and boundary conditions, addressing implementation challenges, scalability, and generalizability. Finally, Section 5.6 outlines future development pathways, with emphasis on methodological enhancements, inter-organizational applications, and integration with ESG frameworks.

5.1. Theoretical and practical contributions of the organizational technological sustainability assessment model

The empirical application of the Organizational Technological Sustainability Assessment (O-TSA) model demonstrates the efficacy of structured analytics in facilitating informed decision-making in technologically complex and resource-intensive supply chains. The model utilizes a systematic analysis of operational data to generate composite performance indicators, thereby offering a multidimensional perspective on the organization's technological profile. From a descriptive analytics perspective, the assessment quantified each sustainability dimension (Input/Output Availability (IOA), Operational Performance (OP), and Technical Quality (TQ)) through clearly defined indicators, weights, and scoring functions. The resulting Technological Sustainability Index (TSI = 3.93) provided a synthetic snapshot of the organization's current technological maturity.

From a diagnostic perspective, the dimension-specific scores revealed not only systemic strengths (e.g., documentation, quality compliance) but also performance bottlenecks (notably in OP), allowing managers to identify underperforming areas and their likely root causes. This capability corroborates the model's role in facilitating a more profound comprehension of the underlying causes of performance discrepancies, a pivotal characteristic of diagnostic analytics. Recent studies adopting fuzzy-MCDM approaches have similarly demonstrated the importance of structured analytics in diagnosing sustainability bottlenecks across supply chains. For instance, fuzzy-AHP and fuzzy-DEMATEL applications in the mining sector confirmed that technological and operational dimensions are often decisive for sustainable performance [55]. Likewise, integrated Delphi-AHP-CoCoSo methods applied to food supply chains have highlighted the critical role of technological efficiency in balancing resilience with sustainability objectives [56]. These results are consistent with the O-TSA findings, particularly in emphasizing the diagnostic capability of analytics to identify systemic weaknesses that can compromise long-term resilience.

The second research question (RQ2) focused on longitudinal improvement, which was addressed through the Technology Improvement Index (TII). The observed TII of 12.93 % over a one-year period

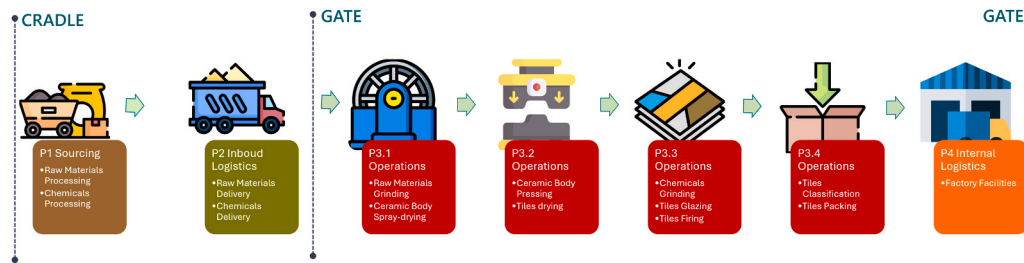


Fig. 5. Representation of the cradle-to-gate value chain of ceramic production [54]. The figure illustrates the supply chain configuration for ceramic production, covering raw material extraction, production stages, and logistics. By situating the Organizational Technological Sustainability Assessment framework within this broader supply chain context, the figure emphasizes how organizational technologies are embedded in systemic interdependencies that influence overall sustainability performance.

validated the model's capacity to capture change over time, monitor progress, and detect strategic inflection points. This dynamic perspective contributes to the domain of prescriptive analytics by providing evidence-based prioritization to technology roadmaps and investment strategies.

5.2. Integration with existing sustainability frameworks

The O-TSA model is intended to complement, rather than replace, existing sustainability assessment frameworks. In line with what recent research has proposed [57–59], which underscores the role of Digital Product Passports and multi-criteria methodologies in integrating digitalization and sustainability into corporate strategies, the O-TSA framework adopts a structured and data-driven approach to technological sustainability assessment, enabling robust data governance and transparency across the supply chain. This alignment with emerging digital enablers positions O-TSA as a practical tool for organizations navigating the twin transition toward digitally integrated and sustainable industrial ecosystems. Beyond complementarity, the integration of technological sustainability introduces both synergies and potential trade-offs with environmental and social objectives. For example, investments in energy-efficient equipment simultaneously enhance technological and environmental sustainability, while extended technology lifespans improve both resource efficiency and organizational resilience. Conversely, rapid digitalization may reduce operational inefficiencies but can generate rebound effects through increased energy demand or electronic waste, creating tensions with environmental goals. Similarly, technological upgrades that improve quality and traceability can strengthen social accountability in supply chains, yet they may also require substantial retraining efforts or cultural adaptation, underscoring trade-offs with the social dimension. By explicitly situating technology alongside the environmental, economic, and social pillars, the O-TSA framework provides a basis for analyzing these interdependencies rather than treating them as exogenous to sustainability assessment.

Conventional approaches, such as Life Cycle Assessment (LCA), primarily focus on environmental impacts, while social sustainability metrics address human and community dimensions. However, not all empirical results converge with our findings. For example, lean-sustainability models in automotive supply chains suggest that operational efficiency improvements often precede gains in technological availability or quality [60]. This partially contrasts with the O-TSA case study, where Input/Output Availability and Technical Quality scored higher than Operational Performance. Such divergences suggest that sectoral context and technological configurations shape the prioritization of sustainability dimensions, reinforcing the need for flexible and industry-specific calibration of the O-TSA framework.

The O-TSA addresses a critical gap in the field by providing a structured methodology for evaluating technological systems. This integration possibility creates a pathway toward a comprehensive sustainability assessment that encompasses all four dimensions:

environmental, economic, social, and technological. The model's compatibility with established frameworks is particularly notable in two areas. Firstly, its foundation in Life Cycle Thinking aligns with the conceptual structure of ISO 14040/14044 standards for environmental assessment, facilitating integration into existing management systems. Secondly, the modular architecture of O-TSA facilitates seamless integration with economic and social indicators in balanced scorecard approaches or sustainability dashboards, potentially enhancing the comprehensiveness of organizational performance monitoring systems. Consistent with recent international evidence, technological sustainability emerges less from isolated improvements and more from the systemic integration of organizational and technological levers, e.g., mass customization routines, digitized workflows, and capability building that co-evolve within a coherent governance architecture. This socio-technical view reinforces the multidimensional logic of the O-TSA: availability (IOA), performance (OP), e quality (TQ) operates as interdependent drivers of resilience, so that partial gains rarely translate into durable, system-level improvement. By explicitly structuring these interdependencies, O-TSA offers an analytics-ready representation that aligns with contemporary approaches to organizational ambidexterity and dynamic capabilities in digital manufacturing.

5.3. Implications for supply chain analytics

For practitioners, the O-TSA provides a concrete method to operationalize technological sustainability and align it with supply chain resilience goals. The structured aggregation of indicators and the visualization of dimension scores enhance cross-functional dialogue, enabling managers to transition from intuitive decision-making to data-driven governance. In contrast to conventional frameworks, such as LCA (which is oriented towards environmental impacts) or lean sustainability models (which are centered on reducing process waste), O-TSA targets the technological foundation of the organization, thereby serving as a complementary and strategic instrument.

The implementation of the O-TSA model in a manufacturing context has yielded several salient analytics-related insights. The implementation of O-TSA underscored the significance of integrating disparate data streams across maintenance, quality, and production systems that had previously been isolated. In addition, the model's three-dimensional structure facilitates the distinction between leading indicators (e.g., documentation completeness and training access) and lagging indicators (e.g., downtime and quality conformance), thereby offering a more nuanced perspective on technological sustainability. Furthermore, the implementation process underscored the significance of contextually relevant threshold setting for the translation of raw measurements into meaningful scores. This finding indicates that supply chain analytics are more effectively informed by industry-specific benchmarking than by universal standards. The radar chart representation of dimensional scores (Fig. 3) proved particularly effective in communicating complex technological sustainability concepts to non-technical stakeholders, reinforcing the importance of visual analytics in supply chain

management.

From a digital transformation perspective, the O-TSA can be embedded within broader supply chain analytics infrastructures. Its modular structure allows seamless integration into digital dashboards that combine financial, environmental, and social KPIs with technological sustainability indicators. In digitally advanced organizations, O-TSA outputs can be synchronized with digital twins for scenario analysis, linked to IoT-based monitoring for continuous updates, and enhanced through machine learning algorithms that detect performance anomalies. This interoperability underscores the dual role of O-TSA: not only as a diagnostic tool but also as a dynamic component of data-driven supply chain governance.

Beyond the technical dimension, the implications of O-TSA also extend to human and organizational dynamics. The effectiveness of technological sustainability is mediated by factors such as user acceptance of new systems, the adequacy of training programs, and the organization's ability to manage change. If these enablers are absent, even robust technological infrastructures may fail to deliver their full sustainability potential. By highlighting such aspects, the O-TSA framework can encourage managers to complement technical interventions with initiatives aimed at workforce development, change management, and cultural adaptation, thereby reinforcing the link between technological and organizational resilience. Recent mixed-method evidence shows that digitalization and "smart practices" are pivotal enablers of circular initiatives, with dynamic capabilities acting as the micro-foundations that channel open innovation into business model innovation and, ultimately, sustainable performance in manufacturing [61].

At the same time, the implications of O-TSA can extend beyond the organizational level. Many technological systems operate within interdependent supply chain networks, where the resilience of one firm depends on the robustness of shared infrastructures, digital interfaces, and supplier technologies. In this context, organizational-level assessments such as O-TSA serve as foundational building blocks for systemic transformation. By providing comparable and transparent measures of technological sustainability, the framework fosters alignment of practices across multiple firms, supports the emergence of collaborative governance mechanisms, and guides joint investment decisions. In this way, O-TSA contributes not only to firm-level resilience but also to the collective sustainability and reconfiguration of industrial networks.

More broadly, these findings resonate with the descriptive and diagnostic layers of supply chain analytics as defined in recent literature. Studies on descriptive analytics in pharmaceutical supply chains have shown that visualization tools and standardized indicators can significantly enhance managerial understanding of sustainability trade-offs [62]. At the same time, diagnostic models in logistics and mining supply chains have revealed that technological gaps often propagate inefficiencies upstream and downstream, echoing the propagation risks identified through O-TSA. By situating technological sustainability within this analytics continuum, the O-TSA framework advances the current debate from isolated performance measures toward integrated, data-driven governance.

5.4. Linking organizational technological sustainability assessment to prescriptive analytics and systemic resilience

Beyond providing descriptive and diagnostic transparency, O-TSA can steer prescriptive choices by translating TSI/TII trajectories into explicit investment rules (e.g., thresholds for retrofits or interoperability upgrades). This role is consistent with recent systematic reviews that frame AI-enabled prescriptive analytics as a lever to strengthen supply chain resilience, moving from prediction to actionable optimization under disruption [63]. Positioning technology analytics within the resilience cycle (readiness, response, recovery, and adaptability) clarifies where organizational technology metrics inform decision policies that accelerate response and shorten recovery windows [64]. From an operations-research perspective, prescriptive analytics encompasses

optimization and decision rules that operationalize resilience strategies, and the literature shows that AI techniques (e.g., fuzzy systems, machine learning, agent-based systems) can act as amplifiers or dampers of risk depending on their governance and integration, hence the need to link O-TSA signals to explicit upgrade/playbooks [65]. Complementarily, reinforcement learning (RL) has matured as a prescriptive paradigm in SCM, particularly for inventory and synchronization decisions, offering adaptive policies that outperform static rules and that can be triggered by O-TSA thresholds when degradation in OP/TQ is detected [66]. Placing O-TSA within the analytics continuum also clarifies how organizational technology metrics feed scenario design, anomaly detection, and policy optimization across tiers. On the scenario side, recent work shows that combining NLP-driven signals with RL supports proactive disruption identification and policy updates, which can be parametrized by O-TSA to focus on the most fragile assets/interfaces [67]. On the simulation/what-if side, digital twin reviews highlight that DTs bolster coordination and resilience (beyond efficiency), enabling prescriptive evaluation of upgrade options (e.g., interface harmonization, redundancy) informed by O-TSA's IOA/OP/TQ deltas [68,69]. Viewed this way, O-TSA evolves from an internal KPI to a shared governance artefact: partners can align standards, compare trajectories (TSI/TII), and coordinate cross-firm upgrades on a common, analytics-ready baseline, an essential lever for systemic change in data-driven supply networks [63,64].

5.5. Limitations and boundary conditions

Despite its demonstrated utility, the O-TSA model has several limitations that warrant explicit acknowledgment. The efficacy of the model is contingent upon the quality and granularity of the available data, which is typically extracted from ERP, MES, and SCADA systems. Organizations with limited digital maturity or fragmented data architectures may encounter difficulties in implementing the full assessment framework without prior investments in data integration and quality assurance. This creates a potential paradox wherein organizations most in need of technological sustainability assessment may be least equipped to implement comprehensive analytics.

In the present application, data availability directly influenced both indicator selection and threshold definition. Several theoretically relevant indicators were excluded because they could not be reliably measured with the company's information systems, while some thresholds were calibrated on external benchmarks when historical series of internal data were insufficient. This pragmatic approach ensured operational feasibility but also introduced dependency on external reference values, which may vary across contexts.

Furthermore, the establishment of thresholds and weights, while grounded in internal benchmarks and expert judgment, introduces interpretive flexibility that has the potential to influence assessment outcomes. The case study employed a combination of Analytical Hierarchy Process (AHP) and expert consensus to establish weights. However, this approach inherently incorporates subjective elements that may vary across organizational contexts. Subsequent implementations would benefit from a more robust weight sensitivity analysis and potentially from machine learning approaches to threshold determination based on performance patterns. More broadly, embedding automation into both weighting and threshold calibration processes would strengthen the objectivity and scalability of the O-TSA. Techniques such as continuous data-driven recalibration, advanced sensitivity testing, or the integration of stakeholder crowdsourcing mechanisms could further mitigate subjectivity and enhance the framework's robustness when applied in diverse industrial environments.

To facilitate implementation in organizations with lower digital maturity, a simplified version of the O-TSA could be adopted, focusing on a reduced set of core indicators complemented by structured surveys or expert elicitation until more robust data systems are established. This staged approach would allow gradual alignment with the full

framework. In practice, such staged implementation can follow a three-phase pathway: (i) a diagnostic phase based on a reduced set of qualitative indicators and structured surveys; (ii) a consolidation phase where digital records from ERP or maintenance systems progressively replace manual data; and (iii) a full integration phase in which automated data pipelines and advanced analytics continuously update the O-TSA indicators. This phased pathway ensures that digitally immature organizations can still benefit from the framework while progressively enhancing their data infrastructure.

Extending the model beyond the ceramic sector requires sector-specific adaptations to ensure both methodological validity and practical relevance. While the three core dimensions of O-TSA (IOA, OP, TQ) retain conceptual coherence, their operationalization must be recalibrated according to industry-specific benchmarks and standards. For example, in automotive manufacturing, Operational Performance (OP) may prioritize reliability and safety-critical indicators, whereas in food processing, Input/Output Availability (IOA) must integrate traceability and compliance with food safety regulations. Such adaptations highlight the importance of methodological flexibility in ensuring the model's broader applicability across supply chain contexts. Scalability also demands attention to organizational size and complexity: for small and medium enterprises (SMEs), the O-TSA can be streamlined by reducing the number of indicators, relying more heavily on expert elicitation, and using simplified scoring schemes that minimize data collection burdens. In contrast, large corporations with diversified technological infrastructures can implement the full O-TSA with extended indicator libraries, sector-specific benchmarks, and automated reporting mechanisms. This differentiation underscores that the framework is not "one size fits all" but rather a modular tool adaptable to organizations of varying scales and resource endowments.

Furthermore, the single-case validation approach, while offering profound insight, restricts the generalizability of findings across diverse industrial sectors and organizational scales. The ceramic manufacturing context is distinguished by its unique technological configurations and sustainability challenges, which may not be directly applicable to other industries. In order to enhance the model's generalizability, multi-case validation across a range of industrial contexts is imperative. Moreover, an overemphasis on organizational boundaries may hinder critical technological sustainability considerations at the level of supply chain networks. Technologies are becoming increasingly pervasive, transcending organizational boundaries and fostering interdependencies that influence overall supply chain resilience. Subsequent extensions of the O-TSA model should consider inter-organizational technological interfaces and network-level sustainability metrics. However, the current application remains constrained to a single-organization perspective. Without explicit network-level indicators (such as interoperability with partner systems, the diffusion of digital technologies across tiers, or dependency on critical suppliers) the model cannot yet capture the cascading effects of technological strengths or weaknesses throughout the supply chain. Expanding O-TSA in this direction would align it more closely with the systemic transformation focus of supply chain analytics research.

Finally, the model's present implementation places significant emphasis on technological performance and quality dimensions, while comparatively neglecting social and cultural factors that influence technology adoption and effectiveness. The human-technology interface, encompassing domains such as user acceptance, skill development, and organizational change management, signifies a pivotal complementary dimension that future model iterations could potentially incorporate more explicitly. Neglecting these factors risks underestimating the systemic conditions that enable or hinder technological sustainability. For example, user resistance, inadequate training, or insufficient change management may compromise the adoption of technologies that otherwise appear sustainable on technical grounds. Future iterations of O-TSA could therefore integrate explicit human- and organization-centered metrics, such as adoption rates, training

effectiveness, or readiness-for-change indices, to capture the socio-technical dynamics that ultimately shape sustainability outcomes.

5.6. Future development pathways

The study's findings underscore the pivotal interconnection between technological and supply chain sustainability. Technologies must be regarded as nodes of visibility, traceability, and disruption potential; they are not neutral enablers. A weakness in one subsystem may propagate inefficiencies and opacity across the entire value chain. The O-TSA facilitates the transition from a "black box" to a "dashboard" by transforming operational data into structured insight, which is a necessary evolution in the development of resilient, efficient, and analytically managed supply chains.

Additionally, the O-TSA model's architecture fosters adaptability. The scope may be expanded beyond the confines of a single organization to analyze technological sustainability at the network level, thereby enabling assessments across suppliers, partners, or logistics nodes. The adaptability of the framework facilitates integration with sophisticated analytics techniques, such as multi-criteria decision-making (MCDM) methods (e.g., analytical hierarchy process, fuzzy logic) and artificial intelligence-driven data extraction. The utilization of these techniques has the potential to facilitate the implementation of automated threshold setting or continuous monitoring.

A case study was implemented, and its limitations were identified. This analysis suggests several promising development pathways for future iterations of the O-TSA model. The integration of digital twin technologies would facilitate real-time monitoring of technological sustainability and the development of scenario models. The development of machine learning algorithms that are capable of detecting patterns in technological performance data has the potential to automate the identification of improvement opportunities and predict potential vulnerabilities before they manifest as performance issues.

The creation of validated indicator sets for various industrial sectors would facilitate the implementation process while ensuring the maintenance of contextual relevance. In conclusion, extending the model to encompass technological interdependencies across supply chain tiers would facilitate a more comprehensive understanding of systemic sustainability risks and opportunities. The implementation of these development directions has the potential to further enhance the value of the O-TSA model as both an analytical framework and a practical decision-support tool for supply chain managers navigating increasingly complex technological landscapes.

6. Conclusions

This article has introduced and validated the Organizational Technological Sustainability Assessment (O-TSA) model as a novel and analytically grounded framework for evaluating technological sustainability at the organizational level. The study addresses a critical gap in sustainability analysis by focusing on the technological infrastructure that underpins supply chain operations, thereby expanding the analytical toolkit available to both researchers and practitioners navigating industrial transformation.

The empirical application of O-TSA to the ceramic manufacturing sector demonstrated its operational feasibility and revealed key findings. The Technological Sustainability Index (TSI) indicated a medium-high level of maturity, while the Technology Improvement Index (TII) captured a 12.93 % improvement over a one-year period. The assessment highlighted strong capacities in input/output availability and technical quality, while identifying operational performance as the primary area for targeted improvement. These results underscore the model's capacity to generate actionable insights, not only by diagnosing current strengths and weaknesses but also by tracking technological progress over time.

The implications of these findings are particularly relevant for

different categories of stakeholders. For managers and practitioners, the O-TSA provides a structured methodology for translating operational data into strategic insights, enabling informed decisions on technology investments, retrofitting, and maintenance priorities. By explicitly quantifying technological sustainability, the model supports the early identification of hidden dependencies and systemic vulnerabilities, thus reinforcing organizational robustness. For supply chain partners, it offers a shared and transparent framework for evaluating technological strengths and weaknesses across interconnected nodes, fostering collaboration and enhancing overall network resilience. For policy-makers and regulators, O-TSA highlights the importance of incorporating technological dimensions into Environmental, Social, and Governance (ESG) reporting, including compliance with emerging directives such as the CSRD, thereby strengthening the visibility of technology as a driver of sustainability performance.

Among these avenues, the most strategically impactful extension concerns inter-organizational and supply chain-level applications. The systemic nature of technological infrastructures means that vulnerabilities or strengths at one node can propagate across networks, amplifying both risks and opportunities. Embedding O-TSA within collaborative governance mechanisms, supplier evaluation protocols, and inter-firm partnerships would therefore transform the framework from an internal diagnostic tool into a shared platform for collective technological sustainability management, risk mitigation, and coordinated investment across industrial ecosystems. In parallel, the integration of O-TSA with ESG frameworks could fill a critical gap in sustainability assessment, ensuring that technological resilience, adaptability, and innovation capacity are fully incorporated into corporate strategies and regulatory disclosure practices. Beyond compliance, such integration would strengthen strategic governance, enabling organizations to align technological investments with long-term sustainability trajectories and to demonstrate accountability to stakeholders across economic, environmental, social, and technological dimensions.

Ultimately, the O-TSA model represents a significant step forward in making technological sustainability visible, measurable, and manageable within the broader context of supply chains. By transforming fragmented operational data into structured insights, it elevates technological considerations from operational concerns to strategic priorities and provides the analytical foundation to ensure that technological investments deliver lasting value across environmental, economic, social, and technological dimensions. More broadly, its structured metrics can act as a bridge between firm-level evaluations and collective strategies, fostering systemic change toward more digitally integrated, adaptive, and technologically sustainable industrial ecosystems.

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CRediT authorship contribution statement

Davide Settembre-Blundo: Writing – original draft, Supervision. **Luca Iattici:** Resources, Data curation. **Anna Maria Ferrari:** Writing – review & editing, Methodology. **Roberto Rosa:** Validation. **Nicholas Berselli:** Investigation. **Marco Vacchi:** Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Glossary

Analytic Hierarchy Process (AHP): A structured method for organizing and analyzing

complex decisions, based on mathematics and psychology. It uses pairwise comparisons and expert judgment to derive weightings for evaluation criteria.

Cradle-to-Gate: A boundary definition in life-cycle analysis that includes all stages from raw material extraction (cradle) to the point where the product leaves the manufacturing site (gate), excluding distribution, use, and disposal.

Descriptive Analytics: The use of data to understand what has happened in a system. In O-TSA, it supports the quantification of technological performance at a specific time.

Diagnostic Analytics: Analytical techniques aimed at identifying the causes of performance outcomes. In this framework, it reveals inefficiencies or underperforming systems.

Digital Twin: A virtual model that replicates physical systems, enabling real-time monitoring, simulation, and predictive analysis. Referenced in the article as a future extension for O-TSA.

Indicator (KPI): A quantitative metric used to evaluate the performance of a system or process. In O-TSA, indicators are grouped under IOA, OP, and TQ dimensions and are used to compute the composite indices.

Input/Output Availability (IOA): A dimension in O-TSA assessing the availability, integration, and readiness of technological resources necessary for operations, including documentation, interfaces, and training.

Life Cycle Assessment (LCA): A methodology to assess environmental impacts associated with all the stages of a product's life. Serves as a conceptual foundation for LCT and O-LCA.

Life Cycle Thinking (LCT): An approach that extends LCA by encouraging systemic, long-term thinking about impacts, resource use, and sustainability trade-offs across the entire value chain.

Mean Time Between Failures (MTBF): A reliability metric indicating the average time between system or component failures. Used as an indicator in the Operational Performance dimension.

Multi-Criteria Decision-Making (MCDM): A family of methods used to support decisions involving multiple, often conflicting, criteria. The O-TSA model is an example of a structured MCDM application.

Operational Performance (OP): A dimension in O-TSA capturing the effectiveness, efficiency, and stability of technological systems during regular operations (e.g., energy use, downtime).

Organizational Life Cycle Assessment (O-LCA): An extension of LCA methods to

organizations, allowing environmental assessments across all activities. Conceptually aligned with O-TSA, which focuses on technological performance.

Organizational Technological Sustainability Assessment (O-TSA): The proposed framework that evaluates the sustainability of technological systems within a manufacturing organization across three dimensions and two indices.

Prescriptive Analytics: Analytics aimed at recommending optimal courses of action. In O-TSA, used to inform strategic decisions based on performance trends (e.g., TII).

Preventive Maintenance Coverage: An indicator measuring the share of assets that follow a scheduled maintenance plan, used in the IOA dimension to evaluate technological readiness and resilience.

SCADA (Supervisory Control and Data Acquisition): An industrial control system used to monitor and control plant processes in real-time. Used as a data source for O-TSA assessments.

Score Normalization: The process of converting raw indicator values to standardized scores (typically 1–5) based on thresholds, ensuring comparability across dimensions and indicators.

Technical Quality (TQ): A dimension in O-TSA evaluating the robustness, maintainability, and adaptability of technological systems (e.g., conformance, remaining useful life, adaptability).

Technology Improvement Index (TII): A relative metric measuring changes in technological sustainability over time, expressed as a percentage difference in TSI across assessment periods.

Remaining Useful Life: The estimated time span during which a technological asset can continue to perform its intended function effectively before its performance deteriorates below acceptable thresholds or it becomes obsolete due to newer innovations. This indicator captures both wear-out dynamics and the risk of technological obsolescence, and is considered within the TQ dimension.

Technological Sustainability Index (TSI): A composite score representing the overall sustainability of an organization's technological systems at a specific point in time, aggregating scores from IOA, OP, and TQ.

Threshold Classes: Predefined value ranges that determine how raw data is translated into performance scores. These are calibrated on benchmarks, internal targets, or expert judgment.